

DISTRIBUTED ENERGY GENERATION AND STORAGE FOR USE IN GREEN HYDROGEN PRODUCTION

by
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**BACHELOR OF SCIENCE
IN
ELECTRICAL AND ELECTRONIC ENGINEERING**



Department of Electrical and Electronic Engineering
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

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A thesis

submitted as a partial fulfillment of the requirement for the degree of

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CERTIFICATE OF APPROVAL

The thesis entitled as “**DISTRIBUTED ENERGY GENERATION AND STORAGE FOR USE IN GREEN HYDROGEN PRODUCTION**” submitted by **Shaharear Hossain**, bearing Matric ID. **ET223104** of session **Spring 2026**, to the Department of Electrical and Electronic Engineering, International Islamic University Chittagong, has been accepted as satisfactory in partial fulfillment of the requirements for the Bachelor of Science in Engineering degree and approved for the examination held on **5 May 2026**.

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DECLARATION

DECLARATION

We hereby declare that this thesis/project report is the result of my/our own independent work and that no part of this work has been submitted elsewhere for the award of any degree or diploma. This work does not include any contribution from collaborative work, except where explicitly stated in the text.

We further declare that:

Originality and Plagiarism

This report is an original work. All sources of information, ideas, and data have been properly acknowledged and referenced in accordance with the guidelines of the Department of Electrical and Electronic Engineering (EEE), International Islamic University Chittagong (IIUC).

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The data presented in this report are genuine, and have been collected, processed, analyzed, and reported with academic honesty. No data have been fabricated, falsified, or manipulated in any form to influence the results or conclusions.

Human/Animal Subjects *(select as applicable)*

- ☐ This research did not involve human or animal subjects.
- ☐ This research involving human participants was conducted with prior ethical approval from the relevant authority. Informed consent was obtained from all participants, and their confidentiality has been strictly maintained.

Conflict of Interest

The authors declare that there is no conflict of interest that could have influenced the conduct, analysis, or interpretation of this study.

Ethical Compliance

This work has been carried out in accordance with the research ethics guidelines instructed by the department.

Name of the Students	Signature of the Students
Shahrear Hossain	

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Authors

ABSTRACT

The variability of wind and solar power presents major challenges for grid stability and energy management in low-carbon power systems. This paper presents a hierarchical Energy Management System (EMS) for a grid-connected microgrid that integrates wind turbines, photovoltaic (PV) arrays, battery energy storage, and a PEM electrolyzer for green hydrogen (H) generation, while serving a dynamic on-grid power load. A multi-objective optimization framework is developed to minimize grid operational costs and peak demand charges subject to real-time electricity pricing, load profile, and electrolyzer constraints. Using a model predictive control (MPC)-based strategy, the EMS dynamically schedules power flows: during periods of low grid prices or excess wind/PV generation, surplus renewable power is prioritized for H production; during high-price periods or renewable deficits, the battery and grid supply the load. The electrolyzer operates in a flexible load-following mode, providing fast-ramping ancillary services (peak shaving and grid support) without compromising hydrogen output efficiency. Simulation results under realistic wind-solar-load scenarios show that the proposed EMS reduces grid energy imports by 2228% , increases renewable self-consumption to over 87%, and ensures uninterrupted load supply. This integrated power-to-gas approach offers a practical pathway for coupling electricity and hydrogen networks while enhancing grid resilience.

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LIST OF ABBREVIATIONS

CHAPTER 1

INTRODUCTION

1.1 Background

The rapid advancement of Very Large Scale Integration (VLSI) technology has significantly increased the demand for high-performance, low-power, and area-efficient digital circuits. As modern electronic systems continue to become more compact and computationally intensive, optimizing the performance of fundamental digital building blocks has become an essential aspect of integrated circuit design. Among these building blocks, the digital comparator plays a vital role in performing binary data comparison operations, making it an indispensable component in modern digital systems. A one-bit comparator is a combinational logic circuit designed to compare two single-bit binary inputs and determine whether one input is greater than, less than, or equal to the other. The outputs generated by the comparator are commonly represented as $A > B$, $A = B$, and $A < B$. Although the circuit is simple in structure, it serves as the foundation for designing higher-order comparators used in complex digital applications. Digital comparators are widely employed in arithmetic logic units (ALUs), central processing units (CPUs), digital signal processors (DSPs), memory addressing systems, analog-to-digital converters (ADCs), communication systems, and embedded devices. The overall performance of these systems is strongly influenced by the efficiency of the comparator circuit. Therefore, reducing power consumption, propagation delay, transistor count, and silicon area while maintaining correct functionality has become a major research focus in digital VLSI design. Over the years, several logic design techniques have been proposed to improve comparator performance. Conventional CMOS logic provides high reliability and noise immunity but generally requires a larger number of transistors, resulting in increased area and power consumption. To overcome these limitations, alternative approaches such as Pass Transistor Logic (PTL), Gate Diffusion Input (GDI), and hybrid logic techniques have been introduced to achieve better power efficiency, reduced propagation delay, and more compact circuit implementations. This research focuses on the design and analysis of a one-bit comparator using an appropriate logic design methodology. The proposed work aims to investigate its functional behavior and evaluate key performance parameters through simulation, contributing to the development of efficient digital comparator circuits suitable for modern VLSI applications.

1.2 Motivation

The continuous growth of portable electronic devices, embedded systems, and high-speed computing platforms has created an increasing demand for digital circuits that consume less power while maintaining high performance. Since digital comparators are frequently used in arithmetic units, processors, signal processing systems, and data communication applications, improving their efficiency can significantly enhance the overall performance of integrated circuits. Conventional CMOS comparator designs provide reliable operation and good noise immunity; however, they generally require a relatively large number of transistors, which increases silicon area, power consumption, and propagation delay. These limitations become more critical as semiconductor technology scales toward deep submicron processes, where power efficiency

and compact circuit implementation are among the primary design objectives[pxc3892316]. Recent studies have introduced alternative logic design techniques, such as Pass Transistor Logic (PTL), Gate Diffusion Input (GDI), and hybrid logic structures, to address these challenges. These techniques aim to reduce transistor count, minimize power dissipation, decrease propagation delay, and improve overall circuit efficiency while preserving correct logical functionality. Comparative analyses reported in previous research indicate that optimized logic styles can achieve noticeable improvements over conventional CMOS implementations in terms of power, delay, and area.[jetir90391] Motivated by these developments, this research focuses on the design and analysis of an efficient one-bit comparator. The study aims to investigate an optimized circuit implementation and evaluate its functional correctness and performance through simulation. By analyzing key design metrics such as power consumption, propagation delay, and circuit complexity, this work seeks to contribute to the development of compact and energy-efficient comparator circuits suitable for modern VLSI applications.

1.3 Problem Statement

Digital comparators are fundamental combinational circuits used in a wide range of digital systems to compare binary data accurately and efficiently. As semiconductor technology advances, the increasing integration density of modern VLSI circuits demands digital components that consume lower power, occupy less silicon area, and operate with minimal propagation delay. Although conventional CMOS-based comparator designs provide reliable functionality and good noise immunity, they generally require a higher transistor count, leading to increased power consumption, larger chip area, and reduced operational efficiency. These limitations become more significant in portable and battery-powered electronic devices, where energy efficiency is a critical design requirement[pxco389216]. Several alternative logic styles, including Pass Transistor Logic (PTL), Gate Diffusion Input (GDI), and hybrid logic techniques, have been proposed to improve comparator performance. However, each design approach involves trade-offs among power consumption, propagation delay, circuit complexity, transistor count, and implementation cost. Therefore, selecting an appropriate logic style that achieves a balanced improvement in these performance parameters remains a challenging task for VLSI designers [jetr903917]Based on these challenges, this research addresses the need for an efficient one-bit comparator design capable of providing correct logical operation while improving key performance metrics. The study focuses on designing and analyzing the proposed comparator using an appropriate logic technique and evaluating its performance through simulation based on power consumption, propagation delay, and overall circuit efficiency.

1.4 Objectives

The primary objective of this research is to design and analyze an efficient one-bit digital comparator capable of performing accurate binary comparison while achieving improved performance in terms of power consumption, propagation delay, and circuit complexity. The study also aims to investigate the effectiveness of the selected logic design technique through simulation and performance evaluation.

The specific objectives of this research are as follows:

- To study the operating principles and functional characteristics of a one-bit digital comparator.
- To develop the logical design of the proposed comparator using an appropriate logic style.
- To derive and verify the truth table and Boolean expressions of the proposed circuit.
- To implement the comparator at the transistor level using CMOS technology.
- To simulate the designed circuit and verify its functional correctness under different input conditions.
- To evaluate important performance parameters, including power consumption, propagation delay, and overall circuit efficiency.
- To compare the obtained simulation results with those reported in previous studies in order to assess the effectiveness of the proposed design.

The successful completion of these objectives is expected to provide an optimized one-bit comparator design that can serve as a reliable building block for larger digital systems and contribute to the development of energy-efficient VLSI circuits.

1.5 Methodology

This research follows a systematic methodology to design, implement, and evaluate a one-bit digital comparator. The methodology begins with an extensive review of existing comparator architectures and logic design techniques to identify their operating principles, advantages, and limitations. Based on the findings of the literature review, an appropriate logic style is selected to develop the proposed comparator circuit. The design process starts with the development of the comparator truth table and the derivation of the corresponding Boolean expressions. These logical functions are then transformed into a gate-level circuit and subsequently implemented at the

transistor level using CMOS technology. Appropriate transistor dimensions are selected to ensure correct switching characteristics and stable circuit operation. After completing the circuit design, the proposed comparator is implemented in a simulation environment using the selected technology model. Functional verification is performed by applying all possible input combinations to confirm the correctness of the output responses. The circuit is then evaluated under predefined simulation conditions to obtain important performance metrics, including power consumption and propagation delay. Finally, the obtained simulation results are analyzed and compared with those reported in previous studies to assess the effectiveness of the proposed design. This structured methodology ensures that the proposed comparator is evaluated from both

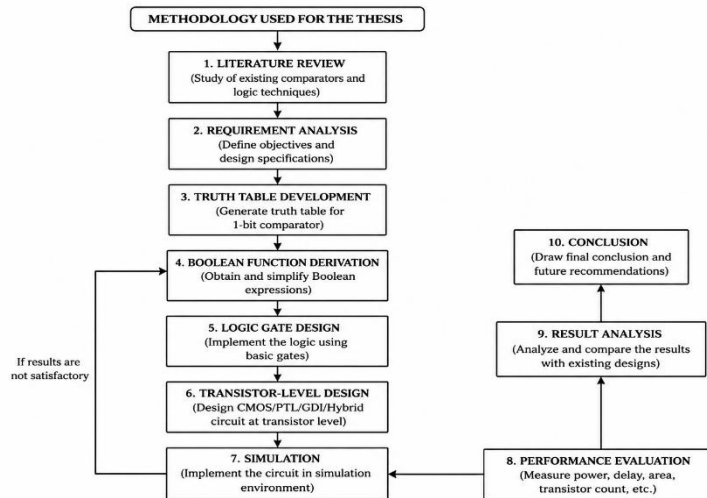


Fig.1.1 Methodology used for the thesis

functional and performance perspectives, providing a reliable basis for validating the research objectives.

1.6 Report Outline

This thesis is organized into five chapters, each focusing on a specific stage of the research. The overall structure is designed to present the research background, theoretical foundation, design methodology, simulation

results, and final conclusions in a logical and systematic manner.

Chapter 1 introduces the research by presenting the background, motivation, problem statement, research objectives, methodology, and the overall organization of the thesis.

Chapter 2 reviews the existing literature related to digital comparators. It discusses the fundamental concepts of comparator circuits, conventional CMOS design, and previous research based on various logic techniques. The chapter also identifies the research gap that motivates the proposed work.

Chapter 3 describes the research methodology adopted in this study. It presents the overall research workflow, system specifications, comparator design methodology, truth table development, Boolean function derivation, logic gate implementation, and transistor-level circuit design.

Chapter 4 presents the simulation results and performance evaluation of the proposed one-bit comparator. The functional verification, short-time and long-time simulation results, system evaluation, and overall performance analysis are discussed to validate the effectiveness of the proposed design.

Chapter 5 concludes the research by summarizing the major findings and contributions of the study. It also discusses the limitations of the proposed work and provides recommendations for future research on efficient digital comparator design.

